
```
n=input('enter a natural number: ');
pf=[];
% create an empty list for the prime factors
check=0;
while true
    if n==1
        disp('empty product')
        % consider the spacial case
        break
    end
    i=2;
    % here, i is the divisor
    % telling function to start the loop from 2 every time
    if n==2
        break
    elseif n==3
        break
    end
    if i<=floor(sqrt(n))
        % the largest amount for the smaller factor is floor of the square root
        for i=2:floor(sqrt(n))
            % here, i is the divisor
            if mod(n,i)==0
                pf(end+1)=i;
                % add i in the list
                n=n/i;
                % divide n by i
                break
            % break for loop and start over from begining but for c/i
            elseif i==floor(sqrt(n))
                check=1;
            end
        end
    end
    if check==1
        break
    end
    if n~=1
        pf(end+1)=n;
        % here adding the last prime into the list
    end
    pf=sort(pf);
    % sort numbers in increasing order
    disp(pf)
    % display the result
```

```
Error using input
Cannot call INPUT from EVALC.
```

```
Error in traditional_division (line 1)
n=input('enter a natural number: ');
```

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